

Model Grid: Transformer Scaling Sweep (L=2, $p = 113$)

Model	d_{model}	Heads	Params	P_{core}	State
M01	56	2	13,946	8,736	—
M02	80	4	28,426	23,104	—
M03	104	4	48,250	43,872	—
M04	128	4	427,253	396,544	Grokked
M05	160	5	656,917	618,560	Running
M06	208	8	1,116,169	1,069,056	—
M07	256	8	1,692,993	1,632,256	—
M08	320	10	2,632,705	2,555,904	—
M09	384	12	3,794,241	3,695,872	—
M10	480	15	5,938,241	5,812,224	—
M11	608	19	9,518,401	9,371,648	—
M12	768	24	15,197,953	14,942,208	—